EE1060 Differential Equations & Transform Techniques
Practice Set 6

This covers Lectures 21—23

Solutions need not be submitted

1. Find the distributional derivatives (i.e., derivative when considered as a distribution) of the following distributions. Remember that $f(t)$ acts as a distribution as follows

$$\langle f, x\rangle = \int_{-\infty}^{+\infty} f(t)x(t)dt.$$

   (a) the 'ramp' function
   $$f(t) = 0 \text{ for } t \leq 0 \text{ and } f(t) = t \text{ otherwise.}$$
   That is, $f(t) = tu(t)$.

   (b) $f(t) = \frac{1}{2}t^2u(t)$

   (c) $f(t) = |t|$ for all t.

   (d) the 'top hat' function
   $$f(t) = 1 \text{ if } |t| \leq 1 \text{ and } f(t) = 0 \text{ otherwise.}$$
   `Remark`: The value of the function at $t = \pm 1$ does not have any impact on the distribution, that is there is no influence on the value of $\langle f, x\rangle$. The solution to this question does not change even if we have $f(\pm 1) = 0$ or $f(\pm 1) = 0.5$.
   `Remark`: When you sketch this function it looks similar to a 'top hat', see below.

Figure 1: Top hat image from Wikipedia: https://commons.wikimedia.org/wiki/File:1910_top_hat.jpg

   (e) $f(t) = \cos(t)u(t)$ where $u(t)$ is the unit step function.

2. Find the impulse response of the following LTI systems. Here x, y denote the input and the output of the system, respectively.

   (a) $y(t) = \int_{\tau=-\infty}^{t} x(\tau)d\tau$

   (b) $y(t) = x(t) + \frac{x(t+1)}{2} - \frac{x(t-1)}{2}$. Also, sketch this impulse response as a 'function' of time t.

   (c) $y(t) = \frac{dx(t)}{dt} - 2\frac{d^2x(t)}{dt^2} + x(t-2)$

   (d) $y(t) = \int_{-\infty}^{+\infty} x(\tau)d\tau$ for all values of t.

3. Find the value of

   (a) $\int_{-\infty}^{+\infty} \delta''(t)dt$, interpret this as $\langle \delta'', 1\rangle$, where 1 is the function that takes the value 1 for all t.

(b) $\int_{-\infty}^{+\infty} \delta'(t)dt$, with similar interpretation as above.

(c) $\int_{-\infty}^{+\infty} \delta(t)\exp(j\omega t)dt$

4. A system with impulse response $h_2(t)$ is said to the inverse of a system with impulse response $h_1(t)$ if their cascade has impulse response $\delta(t)$, that is, $h_1 * h_2 = \delta$. Find the inverse of the system with impulse response $h_1(t) = \delta(t - t_0)$.

5. Consider the top hat function $f(t) = \mathbf{1}\{|t| \leq 1\}$, where $\mathbf{1}\{\cdot\}$ denotes the indicator function. Sketch the following

   (a) $f(t)$

   (b) $\sum_{k=-\infty}^{+\infty} \delta(t-3) * f(t)$

   (c) $\sum_{k=-\infty}^{+\infty} \delta(t-1.5) * f(t)$

6. (Problem 2.45 from Oppenheim, Willsky and Nawab) Find the impulse response of a cascaded LTI system consisting of the following two LTI systems in series:

$$y(t) = x'(t) \text{ and } y(t) = x(t) * h(t)$$

   where $h(t)$ is a known signal.

7. As distributions, are $\delta(t)$ and $t\delta(t)$ identical?

8. (Based on Problem 2.64 from Oppenheim, Willsky and Nawab) *Echo Cancellation.*

   Consider an LTI system described by $y(t) = x(t) + \frac{1}{2}x(t-1)$. This describes a channel with 2 paths. Suppose we want to remove the 'echo' $\frac{1}{2}x(t-1)$ by passing y through another LTI system with impulse response $h_2(t)$, so that the output of this second LTI system is $x(t)$.

   Assuming that $h_2(t) = \sum_{k=0}^{+\infty} g_k \delta(t-k)$ determine value of the coefficients g_k such that $y * h_2 = x$.